HIMANSHU KUMAR THAKUR

Embedded Linux | BSP | Robotics Software Engineer

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LinkedIn

PROFESSIONAL SUMMARY

- Embedded Linux & Robotics Software Engineer with hands-on experience in BSP development, Linux bring-up, Device Tree configuration, kernel customization, and hardware-software integration on STM32MP1 and Raspberry Pi platforms.
- Experienced in custom Linux image development using Yocto and Buildroot, board bring-up, peripheral enablement (UART, SPI, I2C, GPIO, PWM), bootloader configuration, and system-level debugging.
- Developed autonomous robotic systems including UGV platforms, robotic arm systems, motor control, encoder feedback, sensor integration, embedded vision, and distributed communication architectures.
- Proficient in C/C++, Embedded Linux, Linux Internals, BSP Development, Device Tree, Kernel Debugging, Device Drivers, IPC Concepts, Multithreading, and System Validation.
- Currently working on robotics and defense-oriented embedded platforms requiring real-time performance, reliability, and end-to-end system integration.

CORE COMPETENCIES

Embedded Linux	C / C++ ,Python	Linux CLI ,CMake
BSP Development	Linux Internals	STM32MP1
Yocto & Buildroot	IPC (Sockets, Shared Memory, Message Queues, DBus)	Raspberry Pi
U-Boot		ATmega
Device Tree (DTS/DTB)	POSIX Threads	UART, SPI, I2C
systemd	Socket Programming	GPIO, PWM
Board Bring-up	Linux Process Management	UGV Development
Kernel Debugging	Multithreading	Robotic Arm Systems

HIGHLIGHTS

- Professional Experience in Robotics & Embedded Systems
- BSP Development and Linux Bring-up on STM32MP1 Platforms
- Custom Linux Development using Yocto & Buildroot
- Developed UGV Platforms, Robotic Arm Systems and Defense-Oriented Embedded Solutions
- Android ↔ Linux ↔ Microcontroller Control & Communication Architectures

PROFESSIONAL EXPERIENCE

Embedded Software Engineer — **May, 2025 – Present**

Aditya Precitech Ltd. , Hyderabad

Worked on Embedded Linux, BSP development, robotics platform integration, and autonomous system development across STM32MP1, Raspberry Pi, and microcontroller-based platforms.

• Engineered and maintained Embedded Linux BSPs for STM32MP1 and Raspberry Pi based robotics platforms.
• Utilized Linux process management, POSIX threads, IPC mechanisms, systemd services, and build systems (CMake/Make) during embedded software development and system integration activities.
• Built and deployed custom Linux distributions using Yocto and Buildroot for robotics and autonomous systems.
• Customized Linux kernel, Device Tree (DTS/DTB), U-Boot, and root filesystem components for hardware-specific deployments.
• Performed board bring-up, peripheral enablement, kernel debugging, and hardware validation activities.
• Implemented communication frameworks between Linux processors and microcontrollers using UART, SPI, and I2C interfaces.
• Integrated BLDC motors, servo motors, encoder feedback systems, sensors, cameras, and actuators into autonomous robotic platforms.
• Designed Android ↔ Linux ↔ Microcontroller control architectures for distributed robotic systems.
• Created operator control panels and monitoring interfaces for robotic arm and defense-oriented platforms.
• Conducted system integration, debugging, performance optimization, system validation, and reliability testing across embedded hardware and software layers.
• Collaborated with electronics, mechanical, and software teams throughout the product development lifecycle.

KEY PROJECTS

Autonomous UGV Platform for Smart Agriculture & Field Automation

Technologies:

Embedded Linux | Raspberry Pi | STM32/ATmega | OpenCV | Computer Vision | Thermal Imaging | AI/ML | Cloud Computing | UART | SPI | I2C | Motor Control

• Designed and developed an autonomous UGV platform for field automation, monitoring, and precision spraying applications.
• Architected a distributed Raspberry Pi ↔ Microcontroller control system integrating sensors, motor drivers, cameras, and cloud-based AI processing modules.
• Integrated 2 camera pipelines (Thermal + RGB) for environmental monitoring and target detection, and image acquisition.
• Developed cloud-connected workflows for transmitting field data to AI/ML models and receiving autonomous control decisions.
• Implemented precision spraying operations based on AI-generated commands and sensor-driven decision logic.
• Integrated motor control, encoder feedback, ultrasonic sensors, and actuator systems for navigation and task execution.
• Performed system integration, debugging, real-time validation, reliability testing, and field deployment under real-world operating conditions.

STM32MP1-Based Robotic Arm Platform

Technologies:

STM32MP1 | Embedded Linux | BSP | Yocto | Buildroot | Device Tree | Motor Control | Encoder Feedback | Stereo Vision | OpenCV

- Built a custom BSP and Embedded Linux stack for an STM32MP1-based robotic arm platform.
- Customized Device Tree (DTS/DTB), kernel configuration, and peripheral interfaces including GPIO, PWM, UART, SPI, and I2C.
- Enabled hardware interfaces, validated peripheral functionality, and resolved low-level integration issues during STM32MP1 platform deployment.
- Integrated motor drivers, encoder-assisted feedback systems, and joint-control mechanisms to achieve precise robotic arm articulation and trajectory control.
- Engineered calibration and visualization interfaces for robotic arm operation, motion tuning, and system diagnostics.
- Integrated stereo camera systems for object localization, depth estimation, and robotic manipulation workflows.
- Implemented object detection and vision-assisted control workflows to support autonomous pick-and-place operations.
- Calibrated motion-control subsystems, optimized vision-assisted workflows, and verified robotic arm performance under real-time operating conditions.

MMTS & RIS – Defense Target Intervention System

Technologies:

Android (Kotlin) | Raspberry Pi | ATmega |ESP32 | Motor Control | UART | IR Camera

- Architected a distributed control system connecting Android applications, Raspberry Pi controllers, and microcontroller subsystems.
- Built a Kotlin-based Android control application for centralized monitoring, operation, and automation of target intervention systems.
- Established Android ↔ Linux ↔ Microcontroller communication pipelines for real-time command execution and status monitoring.
- Engineered precision motor-control and synchronization algorithms for automated target movement systems.
- Implemented synchronized target-tracking and actuation workflows to achieve accurate target movement and positioning during automated training operations.
- Designed calibration dashboards and visualization tools for robotic arm tuning, diagnostics, and operational monitoring..
- Validated end-to-end communication, command execution, and control-system responsiveness across software and hardware layers.
- Enhanced communication stability, command accuracy, and overall system responsiveness through systematic testing and optimization.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science & Engineering
Siddharth Institute of Engineering & Technology , Andhra Pradesh
Graduated: May 2025

LANGUAGES

- **Hindi**
- **English**